

JINDAL STAINLESS LTD

Steel Melt Shop | AOD Operations

AOD Auto Batching

Business Requirements and Process Mapping Document

Portfolio sample based on operational process improvement work

Document Type	Business Requirements / Process Mapping / Operational Control
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Role Context	Operations Analyst performing Business Analysis activities
Project Area	AOD Auto Batching and Material Addition Control
Document Status	Portfolio-ready sample work

1. Document Control

Field	Details
Project / Process	AOD Auto Batching
Business Area	Steel Melt Shop, AOD Operations
Document Purpose	Capture business requirements, process rules, controls and validation scenarios for AOD auto batching.
Role Context	Operations Analyst responsible for process analysis, operational documentation and requirement clarification.
Related Operational Inputs	Auto batching step table, implementation requirement note, power failure SOP.
Version	1.0

2. Executive Summary

This document captures the business and operational requirements for the AOD auto batching process used in stainless steel production. The objective was to convert operational rules, material addition logic, feed/discharge rates, colour-coded status behaviour and safety controls into a structured business analysis document.

The work demonstrates practical Business Analyst skills within an Operations Analyst role: requirement gathering, process mapping, business rules documentation, stakeholder coordination, control planning, exception handling and test scenario preparation.

3. Business Background

AOD operations require metallic additions and flux additions during the main blow period. The auto batching process needs accurate material selection, correct bunker configuration, proper feed/discharge rates and clear trigger points based on oxygen phase requirements.

For smooth execution, material shifting, charging hopper movement, feeder condition, belt status, flame protection gates and auto mode readiness must be controlled before and during the process.

4. Business Problem

The operational challenge was to control auto batching steps in a repeatable way so that material additions could be completed within the expected main blow window. Without a structured process view, teams could face ambiguity around bunker priority, weighing hopper usage, colour status interpretation, feeder trips, charging hopper readiness and step table reconciliation.

Problem Area	Business Impact
Material addition timing	Delay in completing main blow additions and risk of process variation.
Bunker and weighing hopper logic	Incorrect priority or simultaneous charging could create operational confusion.
Status visibility	Operators need clear colour-coded status to understand material movement stage.
Charging hopper control	Material must be fully discharged and tare weight must be zero before the next shift.
Safety and exception handling	Feeder trip, AOD tilting and power failure require clear control actions.
Evidence and reconciliation	Step table output should be tallied with AOD discharge table after addition.

5. Business Objectives

- Document a standard process for AOD auto batching execution.

- Define business rules for bunker priority, material shifting, weighing hopper use and charging hopper readiness.
- Improve visibility by documenting colour-coded process status rules.
- Support completion of metallic addition and major flux addition during the main blow period.
- Reduce operational ambiguity between AOD, MHS, Electrical, Mechanical and shift teams.
- Create validation scenarios that can be used for process checks, training and system acceptance.

6. Scope

In Scope	Out of Scope
Auto batching process flow for 200, 300 and 400 series.	Steel grade chemistry design or metallurgical formula ownership.
Feed rate, discharge rate and material quantity requirements.	Procurement or supplier commercial decisions.
Bunker, weighing hopper and charging hopper operating rules.	PLC coding or automation software development.
Colour-coded status interpretation.	Detailed hardware design of conveyor, gates or feeders.
Operational exceptions such as feeder trip, AOD tilting and power failure response.	Final approval authority for plant operating standards.
UAT-style scenarios for process validation.	Financial business case beyond process improvement logic.

7. Stakeholder View

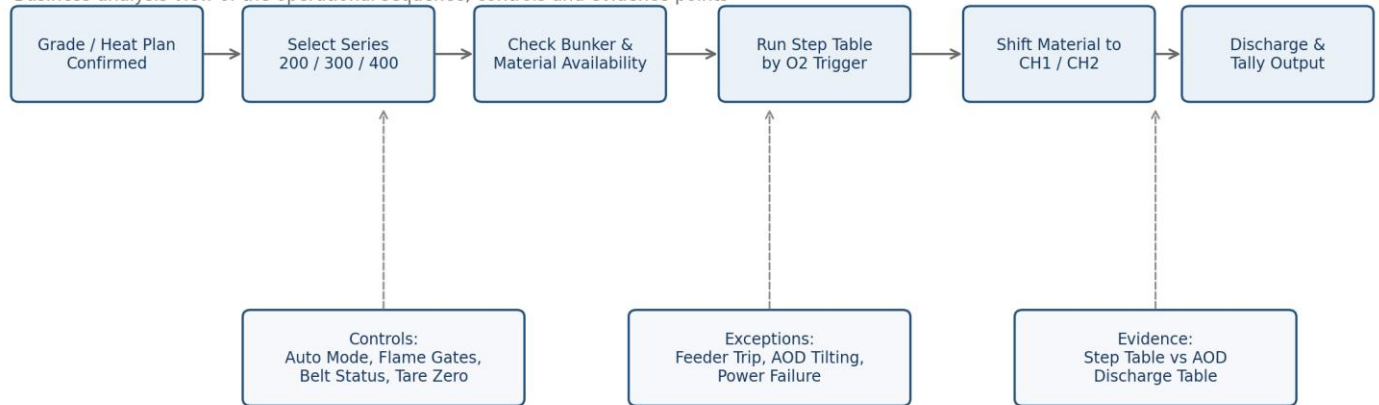
Stakeholder / Team	Role in Process	Requirement Focus
Shift-In-Charge	Owns shift-level execution and decision making.	Clear steps, exceptions, tally checks and safety controls.
AOD Operator	Runs material addition steps and monitors AOD process.	O2 trigger points, charging hopper readiness and discharge status.
MHS Team	Responsible for material handling equipment and feeder operation.	Feeder health, belt status, bunker movement and chute availability.
Electrical Team	Supports electrical readiness and flame protection gates.	Auto mode, gate health and power-related exception response.
Mechanical Team	Supports AOD positioning and mechanical issues.	Cooling position, back pressure checks and emergency handling.
Production Management	Tracks productivity and process performance.	Main blow timing, process discipline and shift compliance.
Quality / Process Team	Checks grade requirements and output stability.	Controlled TR chemistry and consistency of material addition.

8. End-to-End Process Map

The process map below converts the operational auto batching flow into a BA-style process view. It shows the main flow, control points, exception handling and reconciliation evidence.

AOD Auto Batching End-to-End Process Map

Business-analysis view of the operational sequence, controls and evidence points



9. As-Is and To-Be Process View

Area	As-Is Observation	To-Be Requirement
Process reference	Operational rules were available through notes, step tables and team knowledge.	Maintain a single structured process document for execution and training.
Material movement	Material shifting depended on correct understanding of bunker and weighing hopper rules.	Define priority rules and conditions for simultaneous or sequential bunker operation.
Status understanding	Operators needed to interpret colour-coded states correctly.	Document each status colour and its meaning.
Safety controls	Critical exceptions were handled through SOP and shift experience.	Include exception rules in the process-control section.
Validation	Step table and discharge table comparison was required after addition.	Define reconciliation as a formal post-addition control.

10. Series-Wise Auto Batching Step Tables

The step tables summarise material quantity, O2 trigger, feed rate and discharge time requirements for 200, 300 and 400 series. Times are shown in minutes and are used as operational planning references.

10.1 200 Series

Step	Bunker	O2 Trigger	Material	Qty kg	Feed kg/min	Feed min	Discharge kg/min	Shift min	Remarks
S0	3	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	Shift in 3rd blow of last heat
S0	7	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	
S1	7	100-200	Dolo-lime	3,500	N/A	N/A	0	0	Shift when pouring starts and keep 1.5 MT Dolo on charging chute
S2	11 & 12	1700/2000	HCFeCr	7,000	8,000	0.88	6,000	1.58	Shift as soon as possible if confirmed
S3	1 & 2	2300/2500	DRI	6,000	0	0	2,200	3.73	
S4	9 & 10	3000/4500	HCFeMn	10,000	8,000	1.25	6,000	2.67	
S5	9 & 10		HCFeMn	2,000	8,000	0.25	6,000	1.33	
S6	3	4300/4500	Dolo-lime	3,500	1,200	2.92	1,200	3.92	
Total								13.23	Expected total process time reference

10.2 400 Series

Step	Bunker	O2 Trigger	Material	Qty kg	Feed kg/min	Feed min	Discharge kg/min	Shift min	Remarks
S0	3	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	Shift in 3rd blow of last heat
S0	7 or 8	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	
S1	11 & 12	1700/2000	HCFeCr	7,000	8,000	0.88	6,000	1.58	
S2	1 & 2	2300/2900	DRI	10,000	0	0	2,200	5.55	
S3	9 & 10	2900/3100	HCFeMn	1,500	8,000	0.19	6,000	1.25	
S4	6 or 8	3200/3500	Lime	3,500	1,200	2.92	1,200	3.92	
Total								12.30	Expected total process time reference

10.3 300 Series

Step	Bunker	O2 Trigger	Material	Qty kg	Feed kg/min	Feed min	Discharge kg/min	Shift min	Remarks
S0	3	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	Shift in 3rd blow of last heat
S0	7 or 8	Click OB Disc	Dolo-lime	3,500	N/A	N/A	0	0	

Step	Bunker	O2 Trigger	Material	Qty kg	Feed kg/min	Feed min	Discharge kg/min	Shift min	Remarks
S1	13 or 14	700/1000	FeNi	8,000	6,000	1.33	5,000	1.80	Start shifting when AOD has been blocked
S2	11 & 12	1700/2000	HCFeCr	7,000	8,000	0.88	6,000	2.17	
S3	13 or 14	2400/2800	FeNi	8,000	6,000	1.33	5,000	2.60	
S4	1 & 2	2900/3200	DRI	4,000	0	0	2,200	2.82	
S5	9 & 10	3200/3600	HCFeMn	2,500	8,000	0.31	6,000	1.42	
S6	6 or 8	3700/4000	Lime	3,500	1,200	2.92	1,200	3.92	
Total								14.72	Expected total process time reference

11. Functional Requirements

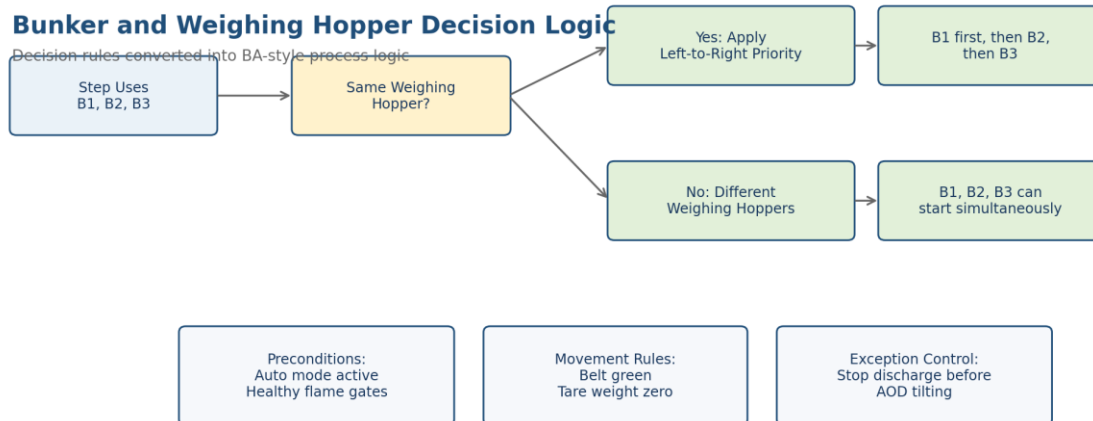
Req ID	Area	Requirement	Priority
FR-01	Series Selection	System/process shall support auto batching step tables for 200, 300 and 400 series.	High
FR-02	Material Setup	Process shall define material name, quantity, bunker number, feed rate and discharge rate for each step.	High
FR-03	O2 Trigger	Each material addition step shall be linked to a defined O2 trigger or process event.	High
FR-04	Colour Status	Process shall display or follow colour-coded status for filling, shifting, ready/discharging and completed states.	High
FR-05	Bunker Priority	If multiple bunkers use the same weighing hopper in one step, left-to-right priority shall be followed.	High
FR-06	Parallel Bunker Use	If bunkers use different weighing hoppers in one step, simultaneous start shall be allowed.	Medium
FR-07	Charging Hopper Control	Material shall shift only after previous material is fully discharged and tare weight is zero.	High
FR-08	Belt Status	Material shall shift only when belt status is green; red status shall block shifting.	High
FR-09	Auto Mode	Equipment required for auto batching shall remain in auto mode unless exception handling is active.	High
FR-10	Post-Addition Check	After addition, step table shall be tallied with AOD discharge table.	High
FR-11	Tilting Control	During AOD tilting in main blow, both charging hoppers shall be discharge stopped before tilting.	High
FR-12	Emergency Handling	During power failure or critical interruption, additions shall stop and emergency checks shall be initiated.	High

12. Business Rules

Rule ID	Business Rule
BR-01	Metallic addition and major flux addition should be completed within the main blow planning window.
BR-02	Bunker 3 shall be allocated for Dolo-lime as part of the auto batching configuration.
BR-03	Selected bunkers not required for a specific grade family shall be marked out of service.
BR-04	Light Green status means weighing hopper filling is in progress or filling not yet completed.
BR-05	Yellow status means material is shifting to charging chute.
BR-06	Orange status means all material has shifted to CH1 or CH2.
BR-07	Dark Blue status means material is ready to discharge or discharging.
BR-08	Sky Blue status means the step is completed.
BR-09	For same weighing hopper, bunker priority shall follow left-to-right sequence and shall not start simultaneously.
BR-10	For different weighing hoppers, bunkers can start simultaneously within the same step.
BR-11	Feeder shall not trip during charging or discharging; MHS team owns feeder readiness.
BR-12	Outlet flame protection gate must be healthy and inlet flame protection gates must be manually open where required.
BR-13	After zero batch is shifted to CH1 or CH2, the empty source shall be marked out of service.
BR-14	In case of main blow power failure, top lance emergency shall be pressed so the top lance returns to rest position.

13. Bunker and Weighing Hopper Decision Logic

The decision logic below converts operational bunker rules into a clear process decision view. This helps operations and automation stakeholders understand when materials should be shifted sequentially or simultaneously.



14. Data Requirements

Data Field	Purpose	Example
Series	Identifies applicable grade family step table.	200 / 300 / 400
Step	Defines sequence number for batching step.	S1, S2, S3
Bunker No.	Identifies source bunker for material.	11 & 12
O2 Trigger	Defines process event or oxygen trigger for addition.	1700/2000
Material Name	Defines material to be added.	HCFeCr, DRI, FeNi
Material Quantity	Defines target material quantity.	7,000 kg
Feeding Rate	Defines material feeding rate.	8,000 kg/min
Discharge Rate	Defines belt discharge rate.	6,000 kg/min
Time to Shift Chute	Calculates expected shift time.	1.58 min
Remark	Captures execution condition or operational note.	Start shifting when AOD has been blocked

15. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Safety	Process must prevent unsafe material shifting during AOD tilting, power failure or equipment abnormality.
NFR-02	Usability	Status colours and step tables must be easy for operators and Shift-In-Charges to interpret.
NFR-03	Traceability	Step table and AOD discharge table must be reconcilable after addition.
NFR-04	Reliability	Feeder, gates and auto mode readiness must be checked to reduce process interruption.
NFR-05	Maintainability	Business rules should be easy to update when grade, bunker or material configuration changes.

16. Exception Handling and Control Plan

Exception	Immediate Control	Owner / Support	Evidence
Feeder trip during charging/discharging	Stop, check feeder health and resolve before continuing material movement.	MHS Team	Equipment status and shift log
Belt shows red status	Do not shift material until status is green.	AOD Operator / MHS	Status indication
Material not fully discharged from CH1/CH2	Do not shift next material until discharge is complete and tare weight is zero.	AOD Operator	Tare weight check
AOD tilting during main blow	Both charging hoppers must be discharge stopped before tilting.	Shift-In-Charge / AOD Operator	Operator confirmation
Power failure during blowing	Stop all additions, confirm Argon emergency valve, call Electrical and Mechanical teams, move AOD to cooling position as required.	Shift-In-Charge / Electrical / Mechanical	Power failure response log
Main blow power failure	Press top lance emergency so the top lance returns to rest position.	AOD Operator	Emergency action confirmation

17. Validation / UAT-Style Scenarios

Scenario	Precondition	Action	Expected Result
UAT-01	200 series selected	Run 200 series step table	System/process follows S0 to S6 sequence and expected total time is around 13.23 minutes.
UAT-02	300 series selected	Run 300 series step table	FeNi, HCFeCr, DRI, HCFeMn and Lime steps follow the defined sequence.
UAT-03	Same weighing hopper used by multiple bunkers	Trigger B1, B2 and B3 in same step	B1 starts first, then B2 and B3; simultaneous start is blocked.
UAT-04	Different weighing hoppers used	Trigger B1, B2 and B3 in same step	All allowed bunkers can start simultaneously.
UAT-05	Belt status red	Attempt material shift	Material shift is not allowed.
UAT-06	CH1/CH2 has material balance	Attempt next material shift	System/process waits until material is completely discharged and tare weight is zero.
UAT-07	AOD tilting during main blow	Initiate tilting control	Both charging hoppers are discharge stopped before tilting.
UAT-08	Power failure during blowing	Apply emergency process	Additions stop, emergency Argon valve is checked and Electrical/Mechanical teams are called.
UAT-09	After addition is completed	Compare step table with AOD discharge table	Material quantities and steps are reconciled.

18. Requirements Traceability Matrix

Requirement	Linked Rule / Control	Validation Scenario	Business Value
FR-01	Series-wise step tables	UAT-01, UAT-02	Supports grade-wise process mapping
FR-02	Material setup table	UAT-01, UAT-02	Ensures quantity and material details are controlled
FR-04	Colour status rules	UAT-05	Improves operator visibility
FR-05	Same WH priority rule	UAT-03	Controls sequential bunker operation
FR-06	Different WH simultaneous rule	UAT-04	Controls parallel bunker operation
FR-07	Tare weight and discharge completion	UAT-06	Prevents incorrect material movement
FR-10	Step table vs discharge table tally	UAT-09	Creates post-addition evidence
FR-11	Tilting control	UAT-07	Controls safety risk
FR-12	Emergency handling	UAT-08	Controls power failure response

19. RACI Matrix

Activity	Shift-In-Charge	AOD Operator	MHS	Electrical	Mechanical	Production
Confirm grade/series step table	A	R	C	I	I	C
Verify material and bunker readiness	A	R	R	I	I	C
Check auto mode and flame gates	A	R	C	R	C	I
Run material shifting/discharge steps	A	R	R	I	I	I
Handle feeder trip	A	C	R	C	C	I
Handle power failure	A	R	C	R	R	I
Tally step table and AOD discharge table	A	R	C	I	I	C

R = Responsible, A = Accountable, C = Consulted, I = Informed

20. BA Skills Demonstrated

BA Skill	How It Is Demonstrated in This Document
Requirement gathering	Captured operational needs around material flow, feed/discharge rates, bunker logic and safety controls.
Process mapping	Converted the auto batching workflow into an end-to-end process map and decision logic.
Business rules documentation	Documented rules for colour status, bunker priority, charging hopper movement, tare weight and exception handling.
Stakeholder analysis	Mapped process responsibilities across Shift-In-Charge, AOD, MHS, Electrical, Mechanical and Production teams.
As-Is / To-Be analysis	Compared informal/operational execution with a controlled, structured process view.
UAT scenario writing	Defined validation scenarios to check step sequence, status logic, bunker rules and emergency handling.
Traceability	Linked requirements to rules, controls and validation scenarios.
Operational change support	Created a professional reference document that supports training, consistency and control.

21. Summary

This document presents the AOD auto batching work as a professional Business Analysis sample. It keeps the original context as an Operations Analyst project and shows the business analysis activities involved in documenting requirements, process flows, business rules, controls, stakeholders, risks and validation scenarios.

This format is suitable for a portfolio sample because it demonstrates how operational process knowledge can be converted into structured requirements and process documentation for digital, automation or business improvement projects.